

Course Code-151603 Ad hoc & Network Sensors

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Unit-1.0: Introduction to Ad Hoc and Sensor Networks**7 hrs**

Characteristics of ad hoc wireless networks, Applications of ad hoc networks, Challenges in ad hoc networks, Wireless sensor networks (WSN) – Characteristics and applications, Design issues in WSN, Sensor network architecture – Layered architecture, Cross-layer design.

Unit-2.0: MAC Protocols for Ad Hoc and Sensor Networks**7 hrs**

Fundamentals of MAC design, Issues in designing MAC protocols, MAC protocols for ad hoc networks – Contention-based protocols (Aloha, CSMA, MACA, MACAW), Scheduling-based MAC protocols, MAC protocols for WSN – S-MAC, T-MAC, B-MAC, X-MAC, Energy-efficient MAC protocols.

Unit-3.0: Routing Protocols in Ad Hoc and Sensor Networks**7 hrs**

Challenges in routing, Proactive vs. Reactive routing, Ad hoc routing protocols – AODV, DSR, DSDV, OLSR, Hybrid routing (ZRP), Hierarchical routing, Geographic routing, Routing protocols for WSN – Flooding, SPIN, LEACH, TEEN, Directed diffusion, Geographic and Energy-Aware Routing (GEAR).

Unit-4.0: Transport and Energy Management in Wireless Sensor Networks**7 hrs**

Issues in transport protocols, TCP over ad hoc networks, Transport protocols for WSN – CODA, RMST, PSFQ, ESRT, Energy-efficient routing in WSN, Data aggregation, Clustering techniques for WSN, Energy harvesting in sensor networks, Sleep-wake scheduling algorithms.

Unit-5.0: Localization, Synchronization, and Data Dissemination in WSN**7 hrs**

Localization techniques – GPS-based localization, Range-based localization (RSSI, TOA, TDOA, AOA), Range-free localization (Centroid, DV-Hop, APIT), Time synchronization in WSN – RBS, TPSN, Data dissemination and query processing in WSN, Mobile sink-based data collection.

Unit-6.0: Security and Emerging Trends in Ad Hoc and Sensor Networks**7 hrs**

Security issues in ad hoc and sensor networks, Attack models – Eavesdropping, Sybil attack, Blackhole, Wormhole, Intrusion detection in WSN, Secure routing in ad hoc networks, Privacy-preserving techniques, Emerging trends – Mobile ad hoc networks (MANETs), Vehicular ad hoc networks (VANETs), Cognitive radio sensor networks (CRSN), AI/ML applications in WSN.

Text/Reference:

1. C. Siva Ram Murthy, B.S. Manoj, Ad Hoc Wireless Networks: Architectures and Protocols, Pearson, 2004.
2. Feng Zhao, Leonidas Guibas, Wireless Sensor Networks: An Information Processing Approach, Elsevier, 2004.
3. Holger Karl, Andreas Willig, Protocols and Architectures for Wireless Sensor Networks, Wiley, 2005.
4. C. K. Toh, Ad Hoc Mobile Wireless Networks: Protocols and Systems, Pearson, 2002.
5. Edgar H. Callaway, Wireless Sensor Networks: Architectures and Protocols, CRC Press, 2003.
6. Thomas Haenselmann, Wireless Sensor Networks: Design Principles for Scalable Distributed Systems, Springer, 2007.
7. Kazem Sohraby, Daniel Minoli, Taieb Znati, Wireless Sensor Networks: Technology, Protocols, and Applications, Wiley, 2007.